Many of these products are covered under other laws, and may not require a WHMIS label and SDS. But if these products are used in your workplace, your employer must still provide you with education and training on their health effects, safe use, and storage.

The rest of this book deals with hazardous products that require WHMIS supplier labels and SDSs (i.e., products covered by WHMIS 2015).

3 Pictograms

About pictograms

What's new

- Pictograms show the type of hazard at a glance.
- There are 10 pictograms.
- Most pictograms have a red, diamond-shaped border.
- Pictograms are assigned to specific hazard classes or categories.

Pictograms are graphic images that immediately show you what type of hazard a hazardous product presents. With a quick glance, you can see, for example, that a product is flammable, or if it might be a health hazard.

Most pictograms have a red, diamond-shaped border. Inside this border is a symbol that represents the hazard, such as fire, health hazard, corrosive, etc. Together, the symbol and the border are referred to as a pictogram. Pictograms are assigned to specific hazard classes or categories.

The following table shows the pictograms. The name of each pictogram is in bold type. The words in the brackets describe the hazard.

Table 4. Introducing the pictograms

 **Exploding bomb** (for explosion or reactivity hazards)	 **Flame** (for fire hazards)	 **Flame over circle** (for oxidizing hazards)
 **Gas cylinder** (for gases under pressure)	 **Corrosion** (for corrosive damage to metals, as well as skin, eyes)	 **Skull and crossbones** (can cause death or toxicity with short exposure to small amounts)
 **Health hazard** (may cause or suspected of causing serious health effects)	 **Exclamation mark** (may cause less serious health effects or damage the ozone layer)	 **Environment*** (may cause damage to the aquatic enviroment)
 **Biohazardous infectious materials**** (for organisms or toxins that can cause diseases in people or animals)		

* There is an environmental hazards group in GHS. This group (and its classes) was not adopted in WHMIS 2015. However, you may see the environmental classes listed on labels and SDSs. Including information about environmental hazards is allowed by WHMIS 2015.

** The "Biohazardous infectious materials" hazard class is included in WHMIS 1988 but is not part of GHS. This class has been kept in WHMIS 2015 to continue to protect workers.

How pictograms compare to WHMIS 1988 hazard symbols



In general, pictograms (at right) are similar to WHMIS 1988 hazard symbols (at left). Many of the symbols inside the borders are almost the same. But there are some new symbols. (See the symbols in the "Health hazard," "Exploding bomb," "Environment," and "Exclamation mark" pictograms, highlighted by dashed lines (above right). And two symbols have been retired. (See the "Materials causing other toxic effects" and "Dangerously reactive material" hazard symbols, highlighted by dotted lines above left). In all but one case ("Biohazardous infectious materials"), the pictogram borders are red and diamond-shaped. And most pictograms are assigned to multiple hazard classes and categories.

For more information

For more information on the meaning of the categories, refer to Annex 1 (Classification and Labelling Summary Tables) of the United Nations publication *Globally Harmonized System of Classification and Labelling of Chemicals (GHS): Fifth revised edition.* unece.org/fileadmin/DAM/trans/danger/publi/ghs/ghs_rev05/English/05e_annex1.pdf

How pictograms are used with WHMIS 2015 hazard classes and categories

The following table shows how the pictograms match up with the hazard classes and categories.

Table 5. Pictograms matched to hazard classes and categories

Pictogram	Hazard classes and categories
	The **flame** pictogram is used for the following classes and categories: • Flammable gases (Category 1) • Flammable aerosols (Categories 1 and 2) • Flammable liquids (Categories 1, 2, and 3) • Flammable solids (Categories 1 and 2) • Pyrophoric liquids (Category 1) • Pyrophoric solids (Category 1) • Pyrophoric gases (Category 1) • Self-heating substances and mixtures (Categories 1 and 2) • Substances and mixtures which, in contact with water, emit flammable gases (Categories 1, 2, and 3) • Self-reactive substances and mixtures (Types B*, C, D, E, and F) • Organic peroxides (Types B*, C, D, E, and F)
	The **flame over circle** pictogram is used for the following classes and categories: • Oxidizing gases (Category 1) • Oxidizing liquids (Categories 1, 2, and 3) • Oxidizing solids (Categories 1, 2, and 3)
	The **gas cylinder** pictogram is used for the following classes and categories: • Gases under pressure (Compressed gas, Liquefied gas, Refrigerated liquefied gas, and Dissolved gas)

Pictogram	Hazard classes and categories
	The **corrosion** pictogram is used for the following classes and categories: • Corrosive to metals (Category 1) • Skin corrosion/irritation — Skin corrosion (Categories 1, 1A, 1B, and 1C) • Serious eye damage/eye irritation — Serious eye damage (Category 1)
	The **exploding bomb** pictogram is used for the following classes and categories: • Self-reactive substances and mixtures (Types A and B*) • Organic peroxides (Types A and B*)
	The **skull and crossbones** pictogram is used for the following classes and categories: • Acute toxicity: – Oral (Categories 1, 2, and 3) – Dermal (Categories 1, 2, and 3) – Inhalation (Categories 1, 2, and 3)
	The **health hazard** pictogram is used for the following classes and categories: • Respiratory or skin sensitization — Respiratory sensitizer (Categories 1, 1A, and 1B) • Germ cell mutagenicity (Categories 1, 1A, 1B, and 2) • Carcinogenicity (Categories 1, 1A, 1B, and 2) • Reproductive toxicity (Categories 1, 1A, 1B, and 2) • Specific target organ toxicity — Single exposure (Categories 1 and 2) • Specific target organ toxicity — Repeated exposure (Categories 1 and 2) • Aspiration hazard (Category 1)

Note

The "Physical hazards not otherwise classified" and "Health hazards not otherwise classified" classes must have a GHS pictogram that matches the hazard identified.

Pictogram	Hazard classes and categories
	The **exclamation mark** pictogram is used for the following classes and categories: • Acute toxicity — Oral, Dermal, Inhalation (Category 4) • Skin corrosion/irritation — Skin irritation (Category 2) • Serious eye damage/eye irritation — Eye irritation (Categories 2 and 2A) • Respiratory or skin sensitization — Skin sensitizer (Categories 1, 1A, and 1B) • Specific target organ toxicity — Single exposure (Category 3)
	The **biohazardous infectious materials** pictogram is used for the following classes and categories: • Biohazardous infectious materials (Category 1)

* Both the flame and exploding bomb pictograms are used for "Self-reactive substances and mixtures" (Type B) and "Organic peroxides" (Type B).

Hazard classes and categories without pictograms

Some hazardous products do not require pictograms. But the product label and Section 2 (Hazard identification) of the SDS still need to show the signal word, hazard statement(s), and other required parts of the label.

The following WHMIS 2015 hazard classes and categories do not need pictograms:

- Flammable gases — Category 2
- Flammable liquids — Category 4
- Self-reactive substances and mixtures — Type G
- Organic peroxides — Type G
- Combustible dusts — Category 1
- Simple asphyxiants — Category 1
- Serious eye damage/eye irritation — Eye irritation — Category 2B
- Reproductive toxicity — Effects on or via lactation

Where to find pictograms

Pictograms will be on the supplier labels of the hazardous products you work with. They will also be on the SDSs, as the symbol or the words that describe the symbol. For more information on labels, see Chapter 4, starting on page 28. For more information on SDSs, see Chapter 5, starting on page 38.

4 Labels

About labels

Under WHMIS 2015, hazardous products used, handled, or stored in the workplace must be labelled. Labels are your first alert about the major hazards of these products. Labels also outline the basic precautions or safety steps you should take.

Overview of label types

There are two main types of WHMIS labels: supplier labels and workplace labels. Other types of identification may be used where appropriate. Examples include warning signs, colour codes, and placards.

WHMIS labels

- All WHMIS hazardous products must be labelled.
- There are two main types of WHMIS labels:
  - Supplier labels
  - Workplace labels
- Other means of identification (for example, warning signs, colour codes, or placards).
- Labels alert you to hazards and safe handling instructions.

What's new

- Supplier labels now require pictograms, signal words, and standardized hazard statements and precautionary statements.
- A pictogram, signal word, and hazard statement are now assigned to most hazard classes and categories.

Supplier labels

A **supplier label** is provided for each hazardous product by the supplier. Supplier labels will appear on all hazardous products received at a workplace in Canada.

Most supplier labels show six types of information. In rare cases, supplier labels show seven types of information.

The written information must be shown in both English and French. Supplier labels may be bilingual (as one label) or available as two labels (one in English, and one in French).

Information required

A WHMIS 2015 supplier label must include the following information:

(1) Product identifier — the brand name, chemical name, common name, generic name, or trade name of the hazardous product.

(2) Initial supplier identifier — the name, address, and telephone number of the Canadian manufacturer or importer.

(3) **Pictogram(s)** — hazard symbol usually contained within a red, diamond-shaped border.
(4) **Signal word** — one of just two words used to alert you to a potential hazard and to state the severity of the hazard. ("Danger" is used for high-risk hazards, while "Warning" is used for less severe hazards.)
(5) **Hazard statement(s)** — a standardized phrase or phrases that describe the type of hazard(s) posed by the hazardous product, as well as its severity. (For example, "Extremely flammable gas," "Fatal if inhaled," and "May cause cancer.")
(6) **Precautionary statement(s)** — a standardized phrase or phrases that describe how to reduce or prevent harmful effects resulting from the following:
   - Exposure to a hazardous product, or
   - Improper handling or storage of a hazardous product

   (Examples include "Keep container tightly closed," "Wear eye protection," and "If exposed or concerned: Get medical attention.")

In rare cases, supplier labels may also include **supplemental label information**. This information gives you more details about the following:

- Precautionary actions
- Hazards not yet included in GHS
- Physical state (gas, liquid, solid, etc.)
- Route of exposure (by inhaling, by swallowing, through skin, etc.)

Supplemental label information is only included on a supplier label in the following cases:

- A toxic mixture has an ingredient with unknown acute toxicity, or
- A product reacts with water to produce an acutely toxic gas

Format

There is no set format for a supplier label, but the pictogram(s), signal word, and hazard statement(s) must be grouped together.

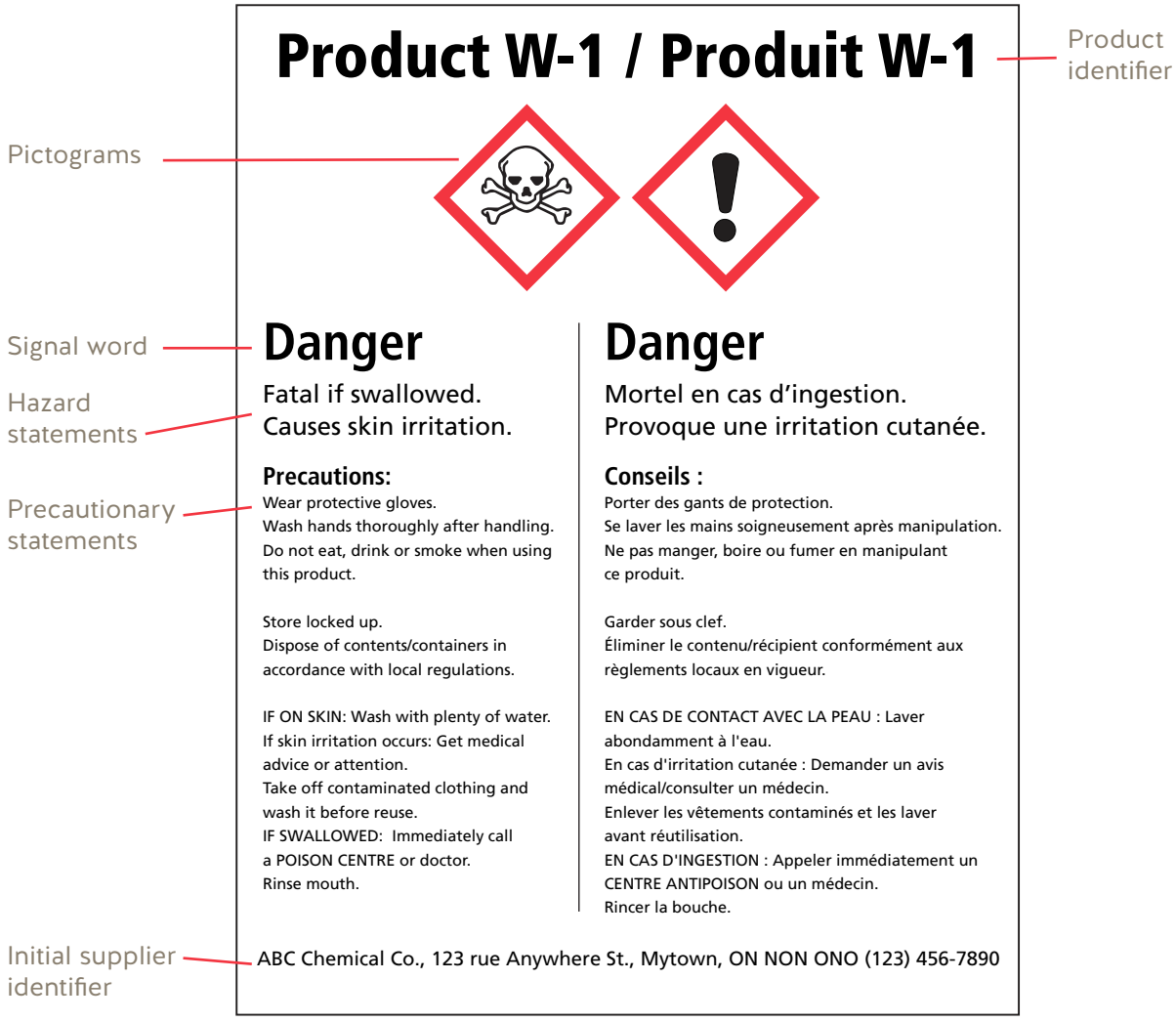
A supplier label must be as follows:

- Clearly and prominently displayed on the container
- Easy to read (i.e., you can see it easily without using any item except corrective glasses)
- In contrast with other information on the product or container
- Bilingual (as one label or two)

If a supplier label becomes damaged, unreadable, or is accidentally removed, your employer must replace the label with either a supplier label or a workplace label.

Sample supplier label

An example of a bilingual supplier label follows.

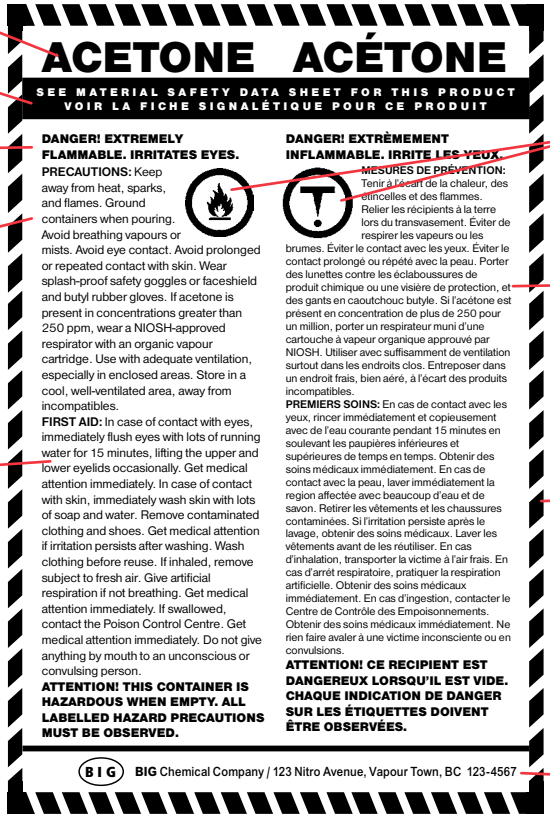


Comparing supplier labels: WHMIS 1988 vs. WHMIS 2015

Two sample supplier labels are shown on the next page so you can compare them: WHMIS 1988 (top) and WHMIS 2015 (bottom).

Comparing supplier labels

WHMIS 1988



Product identifier

Reference to MSDS

Risk phrases

Precautionary statements

First aid measures

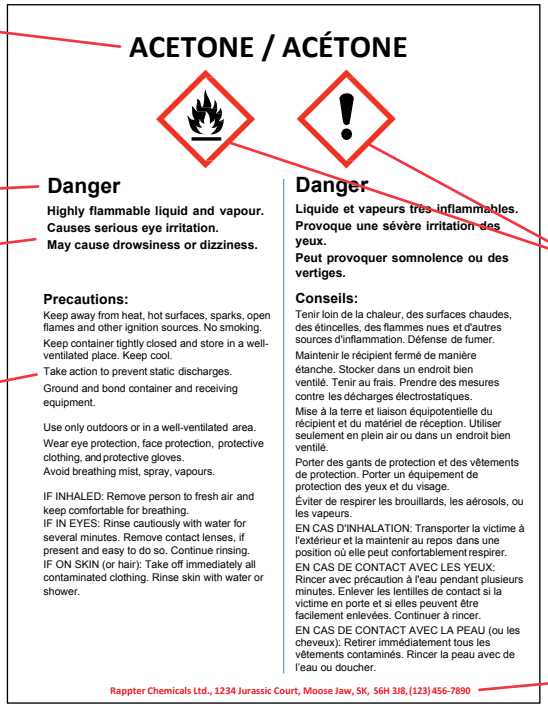
Hazard symbols

French version

WHMIS hatched border

Supplier identification

WHMIS 2015



Product identifier

Signal word

Hazard statements

Precautionary statements

Pictograms

Initial supplier identifier

Workplace labels

A workplace label gives you the following information about a hazardous product:

- The product identifier (name)
- Safe handling information
- A reference to the SDS

Workplace labels may include pictograms or other information from supplier labels.

Format

The format for workplace labels is flexible. For example:

- The information can be written directly onto the container using a permanent marker.
- The wording and language(s) used can be chosen to fit your workplace.

Sample workplace label

An example of a workplace label is shown below:

ACETONE

No smoking, sparks, or flames

Wear eye, face, and hand protection

Use in well-ventilated area, or wear NIOSH-approved respirator with organic vapour cartridges

Safety data sheet available

When workplace labels are needed

A **workplace label** is needed in the following cases:

- A hazardous product is produced (made) and used in your workplace
- A hazardous product is decanted (transferred or poured) into another container
- A supplier label becomes lost or unreadable

In general, your employer is responsible for providing workplace labels. Your employer must also make sure that all labels at your workplace are readable, and that they are replaced if damaged.

Note

Your employer may have rules about labelling that go beyond what's required under WHMIS.

Exceptions for decanted products

Workplace labels are not needed in two specific cases.

The first case is when a hazardous product is decanted from a container that has a supplier or workplace label on it into another container and:

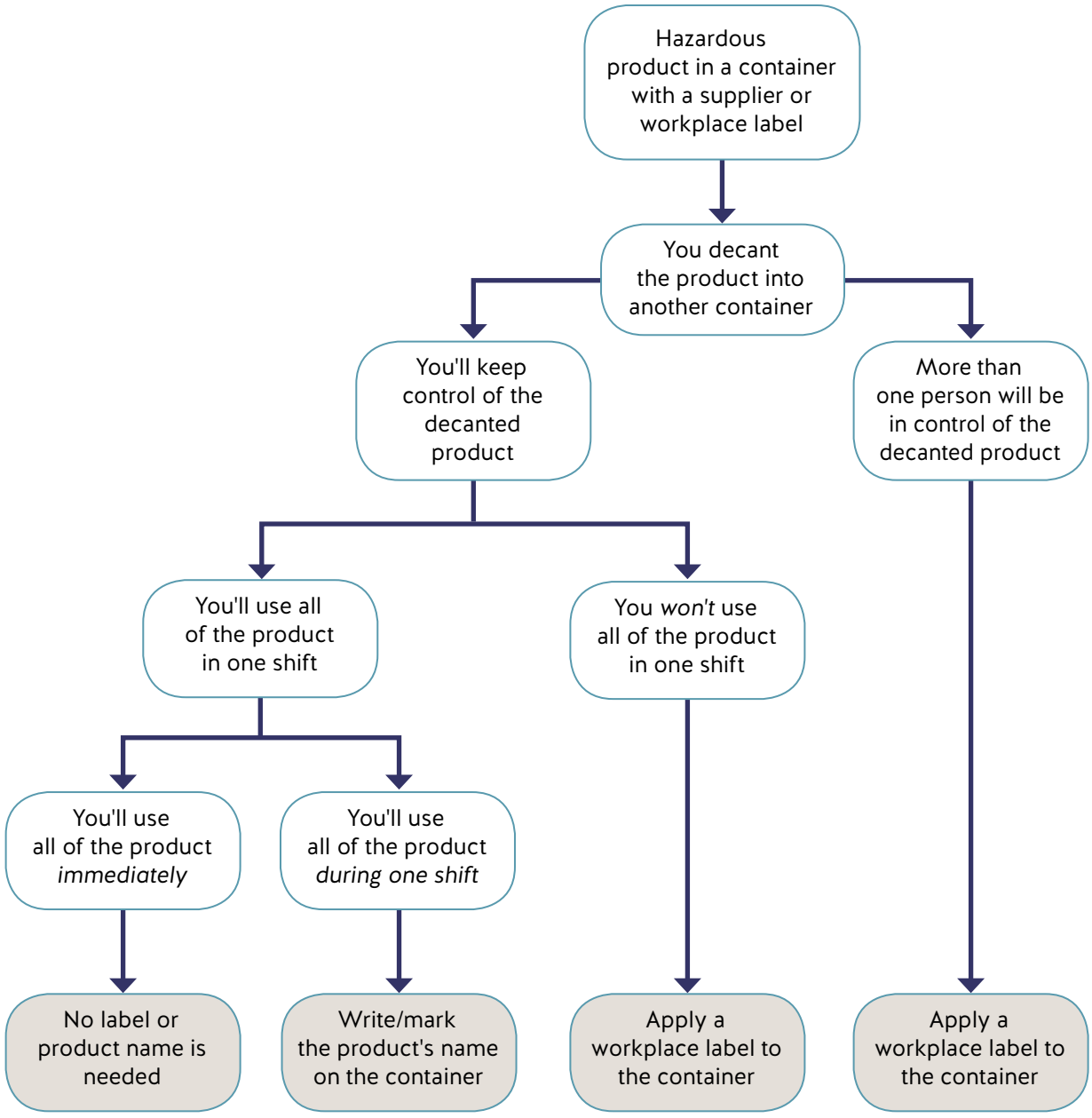
- The decanted product stays under the control of the person who decanted it, and
- The decanted product's name (product identifier) is marked on the container, and
- All of the decanted product will be used during that same shift

For example, if you pour a hazardous product into a container (such as a jar, bottle, or bucket), you write the product's name on the container, you'll be the only person who will use the decanted product, and you'll use all of it during that same shift, then you don't need to apply a workplace label.

The second case is when you'll use the decanted product immediately **and** completely. In this situation, you don't need to apply a workplace label or write the product's name on the container.

The following flow chart walks you through the decision process.

Decision process for labelling decanted products

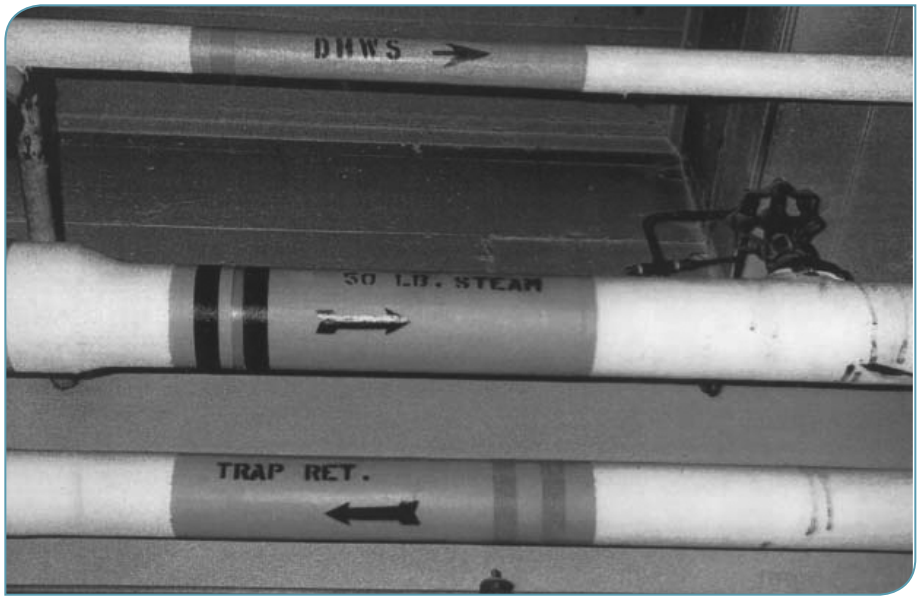


Other means of identification

In some cases, a WHMIS label can also be a mark, sign, stamp, sticker, seal, ticket, tag, or wrapper. It can be attached, imprinted, stencilled, or embossed on the hazardous product or its container. If these other systems are used in your workplace, your employer must make sure you are trained to identify them.

Examples of cases where these other systems may be used, and some exceptions are allowed, include:

- **Bulk shipments** — There is a labelling exemption for products sold without packaging.
- **Small capacity containers, 100 mL or less** — No precautionary or hazard statements are needed on the label.
- **Small capacity containers, 3 mL or less** — The label must be durable and readable, but can be removable for ease of use.
- **Piping systems and vessels** — Labels, placards, colour coding, etc., are allowed.
- **Laboratory samples** — Modified supplier labels are permitted.



Hazardous products in pipes, identified by colours and letters

What you should do when using a hazardous product

As a worker using a hazardous product, you should do the following:

- Always check to see if there is a label on the product before you use it.
- Read, understand, and follow the instructions on the label and SDS. And follow any education, instructions, and training your employer provides.
- Ask your supervisor if you are not sure about how to use or store the product.
- Ask for a new label when you can't see or read the old one properly.
- Do not use a product that is not labelled or if the label is unreadable. Ask your supervisor for help (for example, to replace the label).

5 Safety data sheets

About safety data sheets

What's new

- SDS replaces MSDS.
- Standard 16-section format.
- New information requirements (e.g., WHMIS classification, hazard statements, and other label elements in Section 2).
- SDSs must be accurate at the time of sale or import, for each sale or import.
- SDSs need to be updated when significant new information becomes available.

Safety data sheets (SDSs) are documents that provide information about hazardous products and advice about safety precautions.

An SDS tells you:

- The hazards of a product
- How to use the product safely
- What to expect if you don't follow the advice
- How to recognize symptoms of exposure
- What to do if emergencies occur

SDSs provide more information about products than labels do. SDSs are important resources that help you learn more about the products you use. Use this information to find out the following:

- The hazards of the products you use
- How to protect yourself from those hazards
- Safe handling and emergency procedures

The suppliers of products usually obtain or prepare the SDSs. In some cases, an employer may need to prepare an SDS (for example, when the product is produced and used only in that workplace).

In general, your employer needs to make sure that no SDS is more than three years old. However, there are some exceptions to this rule. For example, if an up-to-date SDS is unavailable, your employer may need to get written confirmation from the supplier that the SDS hasn't changed.

Under WHMIS 2015, every hazardous product that is used, handled, or stored in a workplace must have an SDS.

Uses of an SDS

- A source of detailed information on the hazards of a hazardous product
- An important resource for developing safe work procedures and control measures
- A key part of worker education and training

Rules for completing an SDS

- 16 sections.
- Specific hazardous ingredients must be disclosed. (No “trade secrets” allowed unless a claim has been registered.)
- Any abbreviations used must be defined.
- Information must be specific.
- All required sections must be completed.
- No contradictory information.
- An SDS must be updated within 90 days of significant new information becoming available.

Format and information required

WHMIS 2015 requires a standard 16-section SDS. All information on the SDS must appear in the order shown below.

The table below gives an overview of the information provided in each section.

Table 6. Overview of information required in each section of an SDS

SDS section and heading		Information requirements (partial list)
1	Identification	• Product identifier • Recommended use and restrictions on use • Supplier contact information • Emergency phone number
2	Hazard identification	• Classification (hazard class and category) • Label elements (including pictogram, signal word, hazard statements, and precautionary statements) • Other hazards (for example, heat-related hazards)
3	Composition/ Information on ingredients	For a hazardous product that is a substance: • The chemical name and synonyms • Chemical Abstracts Service Number (CAS No.) • The chemical name of impurities, stabilizing solvents, and stabilizing additives, when required For a hazardous product that is a mixture: the chemical names, synonyms, CAS numbers, and concentrations for ingredients that present health hazards **Note:** Confidential business information (CBI) rules may apply
4	First-aid measures	First-aid procedures by route of exposure (inhalation, skin contact, etc.), as well as the most important symptoms/effects
5	Fire-fighting measures	• Suitable and unsuitable extinguishing media (types of fire extinguishers, such as water, chemical foam, carbon dioxide, etc.) • Specific hazards • Special equipment and precautions for firefighters
6	Accidental release measures	• Protective equipment • Emergency procedures • Methods and materials for containment and cleanup
7	Handling and storage	Precautions for safe handling, conditions for storage, including any incompatibilities

SDS section and heading		Information requirements (partial list)
8	Exposure controls/ Personal protection	Exposure limits, engineering controls, and personal protective equipment
9	Physical and chemical properties	• Appearance • Odour • Odour threshold • pH • Melting/freezing point • Boiling point and range • Flash point • Upper and lower flammable or explosive limits
10	Stability and reactivity	• Reactivity • Chemical stability • Possible hazardous reactions • Conditions to avoid • Incompatible materials • Hazardous decomposition products
11	Toxicological information	Description of various toxic effects by route of entry, including: • Effects of acute (short-term) or chronic (long-term) exposure • Carcinogenicity • Reproductive effects • Respiratory sensitization
12	Ecological information*	• Aquatic and terrestrial toxicity (if available) • Persistence and degradability • Bioaccumulative potential • Mobility in soil
13	Disposal considerations*	Safe handling and methods of disposal, including contaminated packaging
14	Transport information*	UN number and proper shipping name, hazard classes, packing group
15	Regulatory information*	Safety, health, and environmental regulations specific to the product
16	Other information	Other information, including date of the latest revision of the SDS

* Sections 12 to 15 require the headings to be present. The supplier has the option to not provide information in these sections.

Why SDSs can be difficult to understand

SDSs are complex and technical. They are written for many different audiences, including health and safety professionals, employers, supervisors, nurses, doctors, emergency responders, and workers like you.

To make sure that SDS users can quickly find the information they need, information directed toward different users will be listed in specific sections. Having a set format will make it easier to find the information you need on every SDS.

However, you may find some of the information on an SDS difficult to understand. Your employer needs to be able to explain the content of each SDS to you so that you can work safely with or near hazardous products.

When you should use SDSs

Always be familiar with the hazards of a product **before** you start using it.

- Look at an SDS and match the name of the product on the container to the one on the SDS (Section 1).
- Know the hazards (Section 2).
- Understand safe handling and storage instructions (Section 7).
- Understand what to do in an emergency (sections 4, 5, and 6).

You can think of the SDS as having four main purposes. It provides information on the following:

- **Identification** — for the product and supplier
- **Hazards** — physical (fire and reactivity) and health
- **Prevention** — steps you can take to work safely and reduce or prevent exposure
- **Response** — what to do in various emergencies (for example, first aid, a fire, or a spill)

A few things to know:

- Make sure you use the product in the way the manufacturer intended. Otherwise, the advice provided on the SDS and label may not apply, or the safety steps listed may not work. Section 1 of the SDS should describe the typical use of the product and may state restrictions. If the way you use the product does not match the SDS, ask your supervisor or a safety professional for advice.
- Section 2 will sum up the hazards of the product, precautions to take, and what to do in an emergency. But the SDS may not be